

Fuselage Mission Planner

Document: System architecture & complete use-case catalog

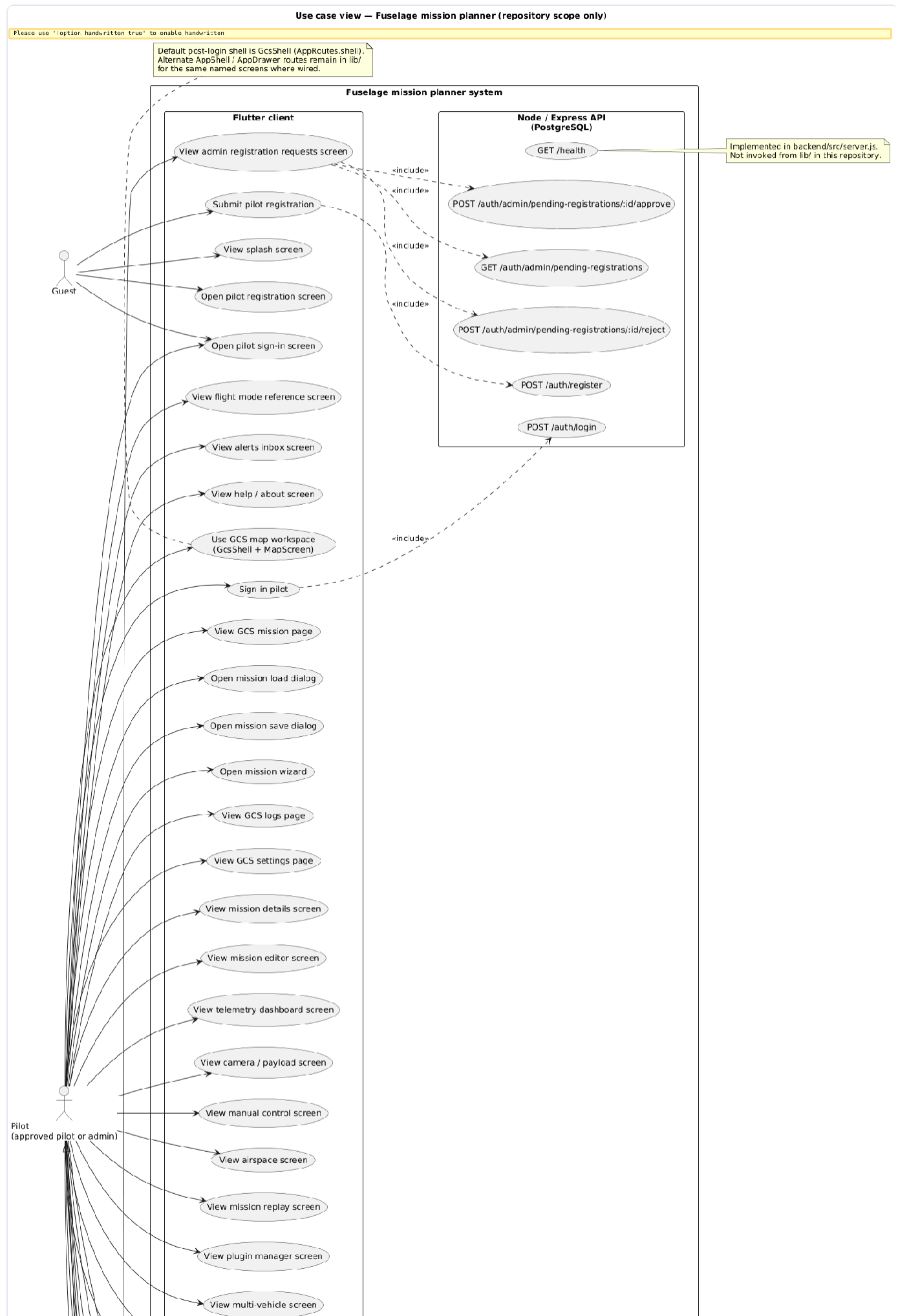
Source: Repository `mission_planner_wireframe` + `backend/`

UML figures: User-supplied PNGs in `docs/arch/` (use case + sequence diagrams)

Generated: see script `build_architecture_document.py`

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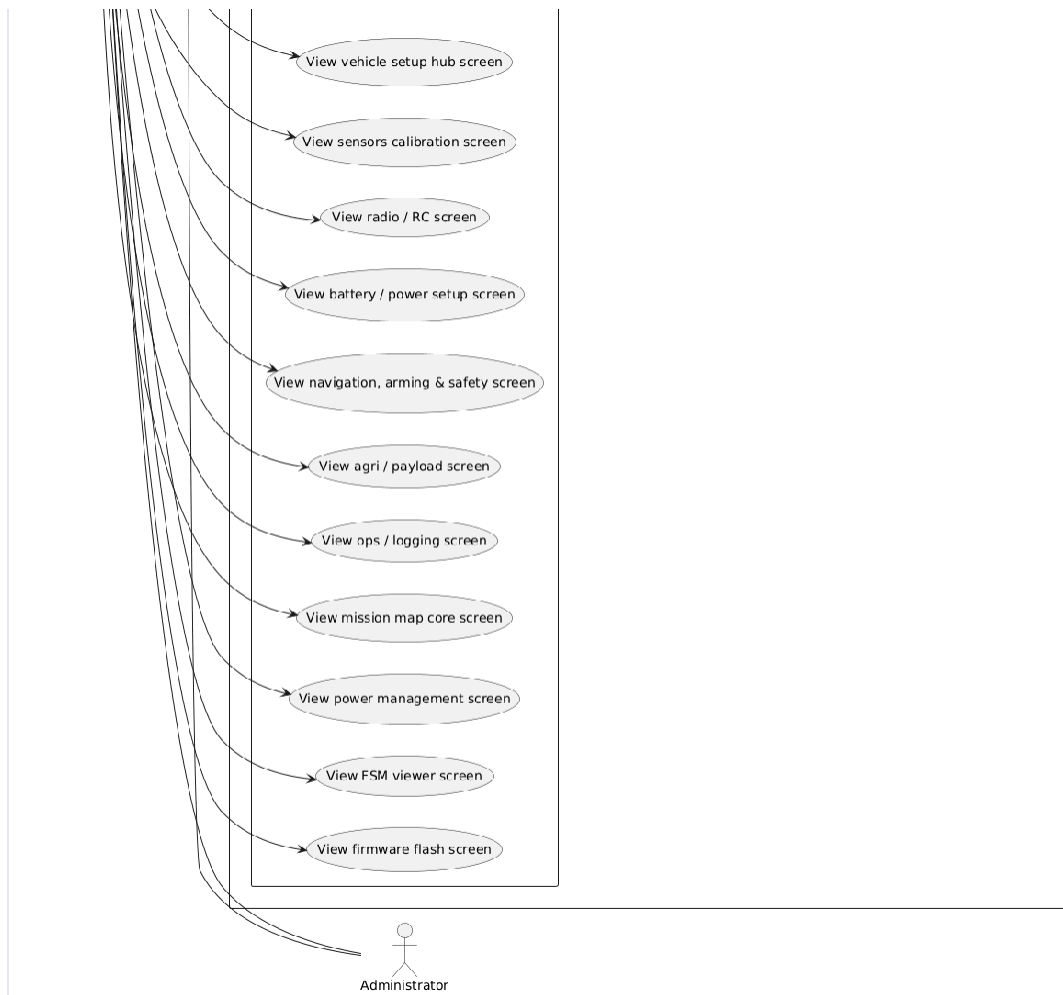


Figure 1 — Use case diagram (Fuselage mission planner system, repository scope)

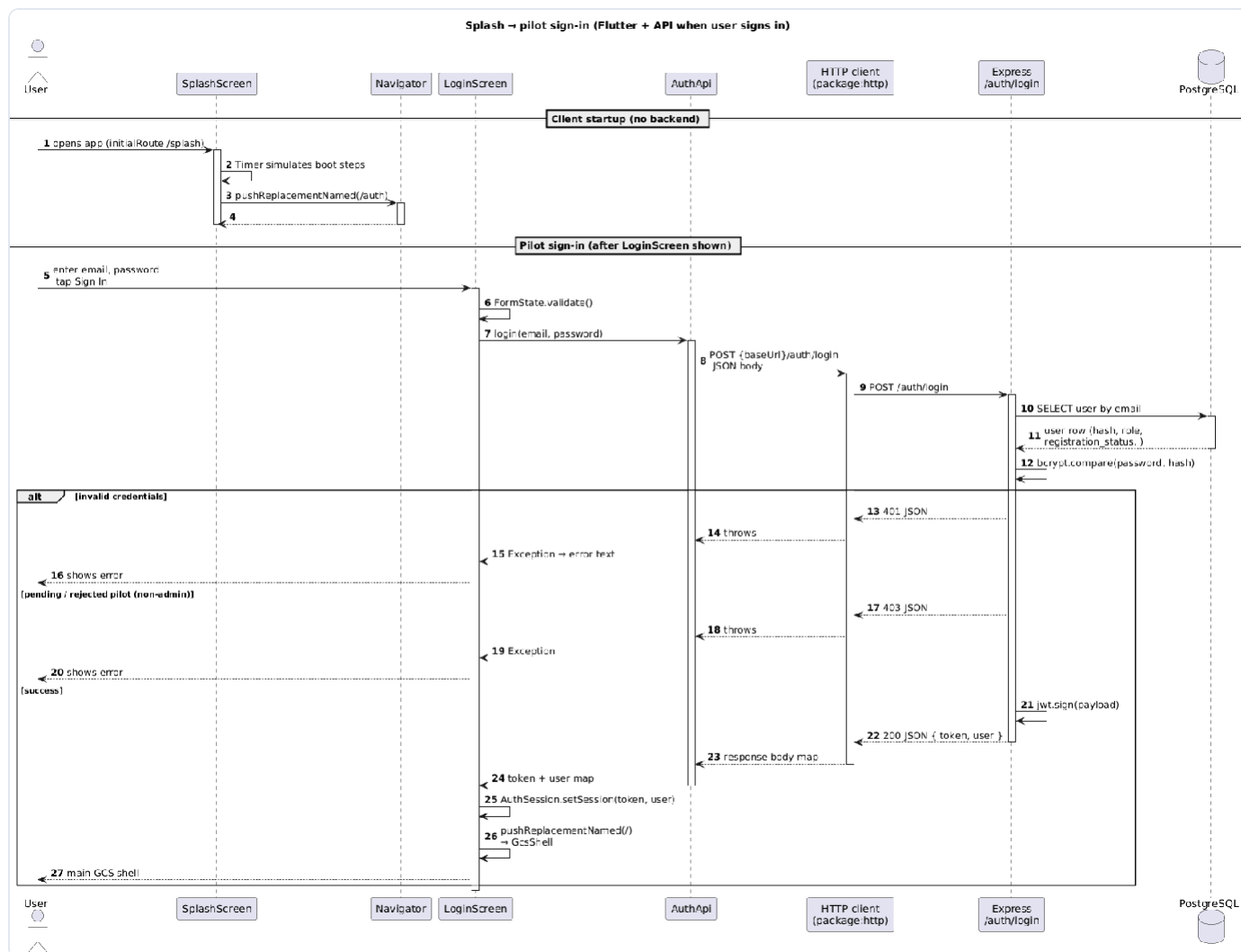


Figure 2 — Sequence diagram: splash → pilot sign-in (Flutter + API)

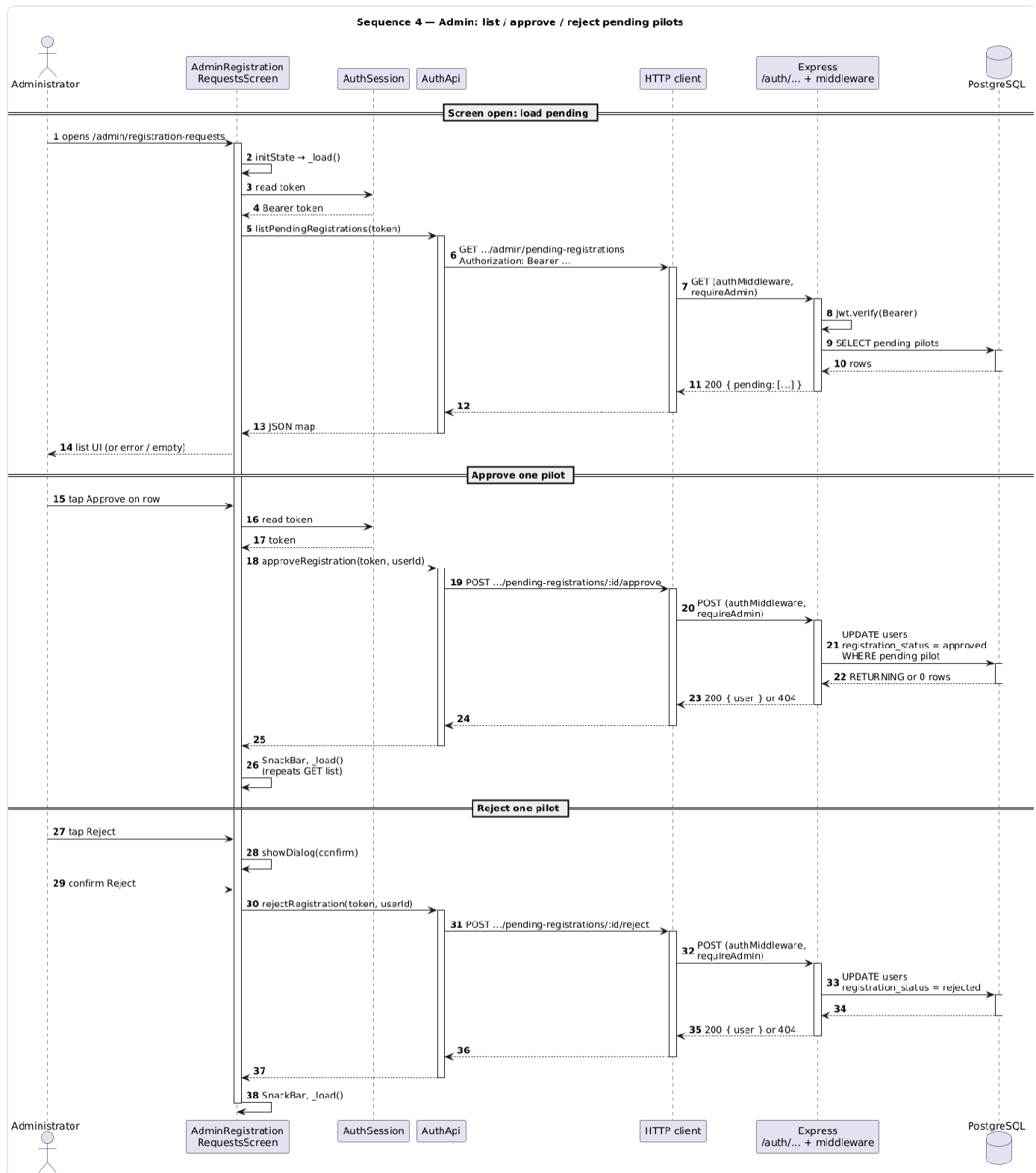
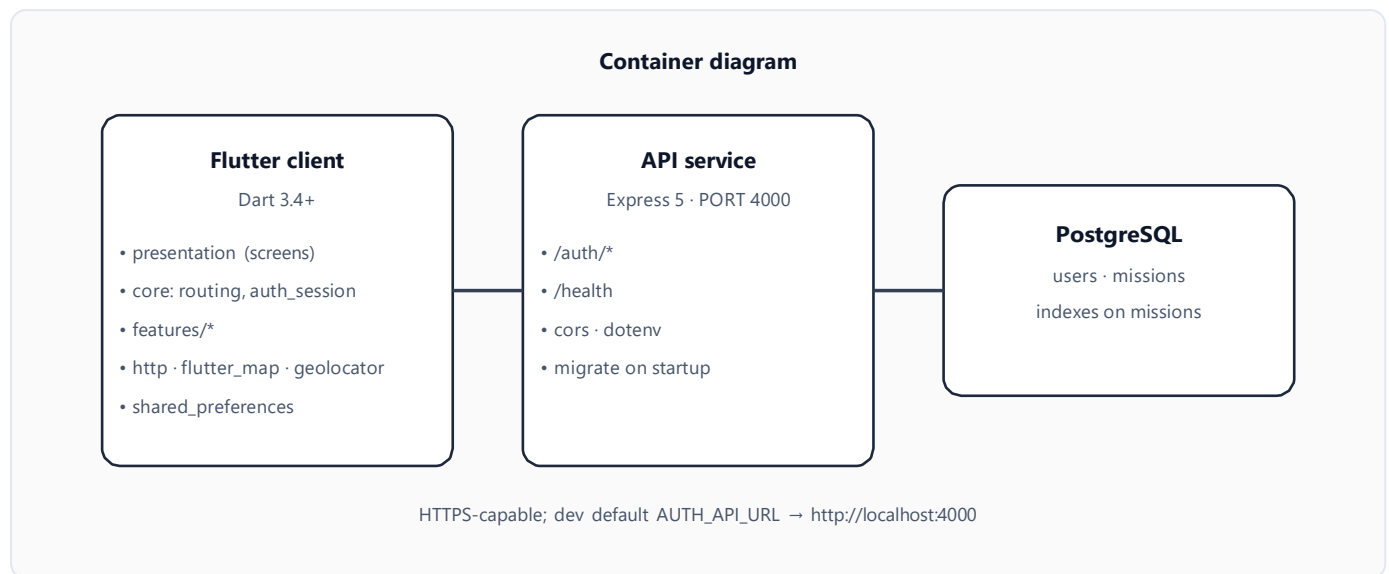


Figure 3 — Sequence diagram: admin list / approve / reject pending pilots

1. System architecture

The following diagrams summarize structure found in `lib/`, `backend/src/`, and `backend/sql/`. They complement the sequence diagrams in §1.

1.1 Containers



1.2 Logical layers

Logical architecture — Flutter packages

UI shell: GcsShell, MapScreen, gcs_top_bar / sidebar / telemetry widgets, MissionPlannerApp

Routing & session: AppRouter · AppRoutes · AuthSession · ThemeController · AppSettingsController

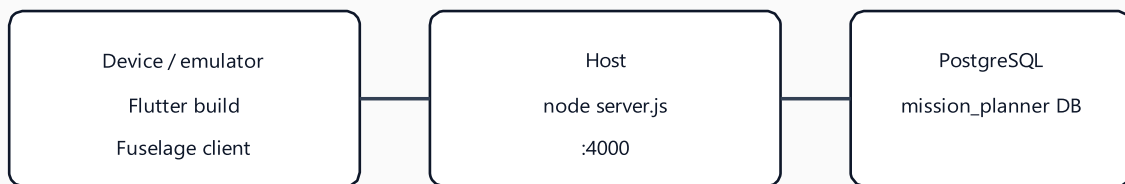
Feature modules: auth, mission, map, dashboard, vehicle_setup, telemetry, replay, ... (presentation + widgets)

Data: AuthApi (http); other features use placeholders / local state / simulated telemetry in-map

Backend stack: server.js mounts express.json, CORS; db.js pool; auth.js router; migrate.js ensures users + missions schema
No REST endpoints for mission CRUD in this repo (missions table exists for future use).

1.3 Deployment (typical)

Typical deployment view (development / small team)

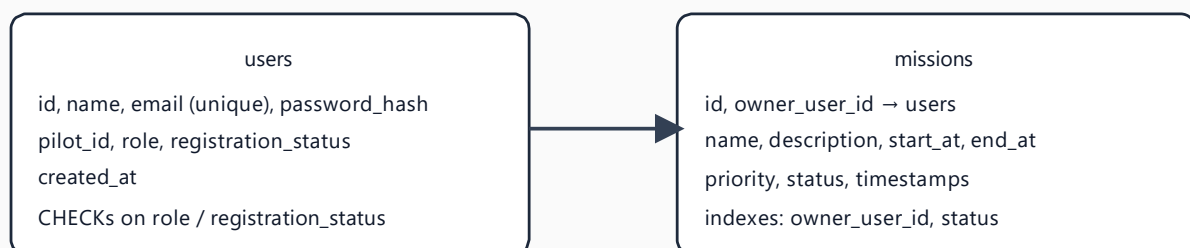


Network: HTTP/JSON between client and API; TLS recommended in production (backend/README).

Map: HTTPS tile requests from device to public tile servers (flutter_map configuration in code).

1.4 Data persistence

Persistent data (PostgreSQL)



2. Complete use-case catalog

Enumerates behaviors implemented in the Flutter router, GCS shell, dialogs, and Express auth/health API.

Admin denotes users with `role == admin` in JWT payload. Alternate `AppShell` / `drawer` routes reuse many of the same named screens but are not the default post-login path (`MissionPlannerApp` uses `GcsShell` after sign-in).

ID	Use case	Primary actor(s)	Stimulus / route / API
UC-001	View splash screen	Guest	<code>`/splash`</code> → <code>SplashScreen</code>
UC-002	Open pilot sign-in screen	Guest	<code>`/auth`</code> after splash; <code>LoginScreen</code>
UC-003	Open pilot registration screen	Guest	<code>LoginScreen</code> → <code>`/register`</code>
UC-004	Submit pilot registration	Guest	<code>RegisterScreen</code> → <code>POST `/auth/register`</code>
UC-005	Sign in pilot	Guest → Pilot	<code>LoginScreen</code> → <code>POST `/auth/login`</code> → <code>AuthSession</code> → <code>`/`</code>
UC-006	Use GCS map workspace	Pilot, Admin	<code>GcsShell</code> + <code>MapScreen</code> (map section)
UC-007	View GCS mission page	Pilot, Admin	<code>GcsShell</code> mission section → <code>MissionPage</code>
UC-008	Open mission load dialog	Pilot, Admin	<code>MissionPage</code> → <code>showLoadMissionDialog</code>
UC-009	Open mission save dialog	Pilot, Admin	<code>MissionPage</code> → <code>showSaveMissionDialog</code>
UC-010	Open mission wizard	Pilot, Admin	<code>`/mission-wizard`</code> (e.g. New mission, map FAB)
UC-011	View GCS logs page	Pilot, Admin	<code>GcsShell</code> logs → <code>LogsPage</code>
UC-012	View GCS settings page	Pilot, Admin	<code>GcsShell</code> settings → <code>SettingsPage</code>
UC-013	Open tools sheet (secondary routes)	Pilot, Admin	<code>GcsShell</code> tools modal → <code>pushNamed</code> routes below
UC-014	View admin registration requests	Admin	<code>`/admin/registration-requests`</code> (+ pending API)
UC-015	View mission details	Pilot, Admin	<code>`/mission-details`</code>
UC-016	View mission editor	Pilot, Admin	<code>`/mission-editor`</code>
UC-017	View telemetry dashboard	Pilot, Admin	<code>`/telemetry`</code>
UC-018	View camera / payload	Pilot, Admin	<code>`/camera-payload`</code>
UC-019	View manual control	Pilot, Admin	<code>`/manual-control`</code>
UC-020	View airspace	Pilot, Admin	<code>`/airspace`</code>
UC-021	View mission replay	Pilot, Admin	<code>`/replay`</code>
UC-022	View plugin manager	Pilot, Admin	<code>`/plugins`</code>
UC-023	View multi-vehicle	Pilot, Admin	<code>`/vehicles`</code>
UC-024	View vehicle setup hub	Pilot, Admin	<code>`/vehicle-setup`</code>
UC-025	View sensors calibration	Pilot, Admin	<code>`/vehicle-setup-sensors`</code>
UC-026	View radio / RC setup	Pilot, Admin	<code>`/vehicle-setup-radio`</code>
UC-027	View battery / power setup	Pilot, Admin	<code>`/vehicle-setup-battery`</code>
UC-028	View navigation, arming & safety	Pilot, Admin	<code>`/vehicle-setup-nav-safety`</code>

ID	Use case	Primary actor(s)	Stimulus / route / API
UC-029	View agri / payload setup	Pilot, Admin	`/vehicle-setup-agri`
UC-030	View ops / logging setup	Pilot, Admin	`/vehicle-setup-ops`
UC-031	View mission map core (setup)	Pilot, Admin	`/vehicle-setup-mission`
UC-032	View power management	Pilot, Admin	`/power`
UC-033	View FSM viewer	Pilot, Admin	`/fsm-viewer`
UC-034	View firmware flash	Pilot, Admin	`/firmware-flash`
UC-035	View flight mode reference	Pilot, Admin	`/flight-mode-reference`
UC-036	View alerts inbox	Pilot, Admin	`/alerts-inbox`
UC-037	View help / about	Pilot, Admin	`/help-about`
UC-038	Resolve unknown route	Pilot, Admin	Unregistered name → `_UnknownRouteScreen`
UC-039	List pending registrations (API)	Admin	GET `/auth/admin/pending-registrations`
UC-040	Approve pilot (API)	Admin	POST `.../pending-registrations/:id/approve`
UC-041	Reject pilot (API)	Admin	POST `.../pending-registrations/:id/reject`
UC-042	Check API and database health	System operator	GET `/health` (not called from `lib/`)

3. Notes & scope

- ♦ **Missions table:** PostgreSQL `missions` exists (see `init.sql` / `migrate.js`); no mission CRUD REST routes are implemented in this repository.
- ♦ **GCS data:** Map and telemetry displays include simulated or placeholder logic unless otherwise wired.
- ♦ **Additional sequences** (e.g. registration POST only) can be documented separately; Figures 2–3 cover splash/login and admin moderation.